

Ryan's SAT Math Drill 2

Instructions

Complete all 22 questions in 35 minutes. A calculator is permitted. Select the best answer for each question. The questions are original but modeled on the skills, structure, and multi-step reasoning of challenging official SAT Math questions.

Questions

1. A linear function f satisfies $f(3) = 17$ and $f(8) = 42$. What is $f^{-1}(67)$?
 - A. 11
 - B. 12
 - C. 13
 - D. 15
2. A laboratory purchased 42 sensors. Basic sensors cost \$48 each, and advanced sensors cost \$75 each. The total cost was \$2,556. How many basic sensors were purchased?
 - A. 18
 - B. 20
 - C. 22
 - D. 24
3. For what value of k does the system below have no solution?
$$(k - 2)x + 6y = 9, \quad 4x + 12y = 25$$
 - A. 2
 - B. 4
 - C. 6
 - D. 8
4. Plan A charges \$65 per month plus \$0.12 per text message. Plan B charges \$35 per month plus \$0.18 per text message. What is the least whole number of monthly text messages for which Plan A costs less than Plan B?

- A. 499
- B. 500
- C. 501
- D. 502

5. The line $Ax + By = 26$ passes through $(3, 2)$ and is perpendicular to $y = \frac{2}{3}x - 7$. What is $A + B$?

- A. 5
- B. 8
- C. 10
- D. 13

6. The polynomial $x^2 + bx + c$ has a zero at $2 + \sqrt{3}$, where b and c are rational constants. What is $b + c$?

- A. -5
- B. -3
- C. 1
- D. 5

7. Which expression is equivalent to

$$\frac{x^2 - 9}{x^2 + x - 6} \cdot \frac{x + 3}{x - 3}$$

for every value of x in the domain of the original expression?

- A. $\frac{x + 3}{x - 2}$, where $x \neq -3, 2, 3$
- B. $\frac{x - 3}{x - 2}$, where $x \neq -3, 2$
- C. $\frac{x + 3}{x - 2}$, where $x \neq 2$
- D. $\frac{x + 3}{x + 2}$, where $x \neq -3, 3$

8. An exponential function $p(t) = ab^t$ satisfies $p(2) = 54$ and $p(5) = 1,458$. What is $p(4)$?

- A. 162
- B. 324
- C. 486
- D. 972

9. A rectangular enclosure borders a straight river, so fencing is needed on only three sides. If 240 meters of fencing is available, what is the greatest possible area of the enclosure?

- A. 3,600 m²
- B. 7,200 m²
- C. 10,800 m²
- D. 14,400 m²

10. For $p(x) = x^3 + kx^2 - 4x + 7$, the remainder when $p(x)$ is divided by $x - 2$ is 11. What is the remainder when $p(x)$ is divided by $x + 1$?

- A. -11
- B. -1
- C. 7
- D. 11

11. The equation $\frac{x+3}{x-1} = k$ has solution $x = 5$. What is the solution of

$$\frac{2x - k}{x + 2} = 3?$$

- A. -8
- B. -4
- C. 4
- D. 8

12. What is the solution of $\sqrt{2x + 15} = x$?

- A. -3
- B. 3
- C. 5
- D. 15

13. The function f has a minimum at $(-1, 4)$. If $g(x) = -2f(x - 3) + 5$, at which point does g have its corresponding maximum?

- A. $(-4, -3)$
- B. $(2, -3)$
- C. $(2, 3)$
- D. $(4, 13)$

14. The graphs of $y = x + 2$ and $y = x^2 - 4x + 8$ intersect at two points. What is the product of the y -coordinates of those points?

- A. 9
- B. 12
- C. 20

D. 25

15. For each value of k , consider the equation

$$x^2 + (k - 1)x + (k + 5) = 0.$$

What is the sum of all values of k for which the equation has exactly one real solution?

- A. -6
- B. -1
- C. 6
- D. 19

16. A container holds 80 liters of a solution that is 35% acid. Some of the solution is removed and replaced by the same amount of pure water. The resulting solution is 28% acid. How many liters were removed?

- A. 12
- B. 16
- C. 20
- D. 28

17. A data set has mean 12 and standard deviation 3. Every value x is transformed to $2x - 5$. What are the mean and standard deviation of the transformed data set?

- A. Mean 7; standard deviation 1
- B. Mean 19; standard deviation 3
- C. Mean 19; standard deviation 6
- D. Mean 24; standard deviation 6

18. Of 120 students, 70 participate in music, 50 participate in a sport, and 30 participate in both. If a student who participates in a sport is selected at random, what is the probability that the student also participates in music?

- A. $\frac{1}{4}$
- B. $\frac{3}{7}$
- C. $\frac{1}{2}$
- D. $\frac{3}{5}$

19. A circle has center $(2, -3)$ and is tangent to the line $y = 5$. When the circle's equation is written as

$$x^2 + y^2 + Dx + Ey + F = 0,$$

what is F ?

- A. -64
- B. -51
- C. 13
- D. 51

20. Two right circular cones are similar. The smaller cone has height 6, and the larger cone has height 15 and volume $1,000\pi$. What is the volume of the smaller cone?

- A. 64π
- B. 160π
- C. 400π
- D. 640π

21. An observer's eyes are 5 feet above level ground. The observer stands 30 feet from a building, and the tangent of the angle of elevation from the observer's eyes to the roof is $\frac{4}{3}$. How tall is the building?

- A. 35 ft
- B. 40 ft
- C. 45 ft
- D. 50 ft

22. In right triangle ABC , $\angle C = 90^\circ$, the hypotenuse has length 26, and $\sin A = \frac{5}{13}$. An altitude is drawn from C to the hypotenuse. What is the altitude's length?

- A. $\frac{60}{13}$
- B. $\frac{120}{13}$
- C. 10
- D. 12